Oh, Hanseok

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Research Interest

I am interested in exploring Wearable and On-device Agents. I believe that for AI to truly mirror human perception, it must move toward multisensory intelligence—leveraging sensors and the "five senses" captured directly from the physical world. Furthermore, I am convinced that agents capable of integrating our digital and physical worlds will become increasingly vital, with embedded agents playing a central role in this evolution.

Education

New York University Jul 2026 – present
PhD in Computer Science (*Advisor*: Muhammad Shafique)
Area: Embedded AI, Wearable & On-device Agent, Multisensory intelligence

Korea Advanced Institute of Science and Technology Sep 2021 – Feb 2024
MS in Artificial Intelligence (*Advisor*: Minjoon Seo)
Area: Natural Language Processing, Machine Learning, Information retrieval

SungKyunKwan University Mar 2014 - Aug 2020
BS in Data Science & BA in Child and Adolescent Education
Cumulative GPA: 4.17/4.5

Experience

Mila @ Montreal, Canada June 2025 - Mar 2026
Visiting Researcher (Advisor: Siva Reddy)
I worked on understanding how current Vision Language Models (VLM) are good at tackling knowledge-intensive question in the wild setting. Currently under review at ECCV 2026.

SoftlyAI @ Seoul, Korea May 2024 - May 2025
AI research Engineer
SoftlyAI is a startup working on the AI Agent for the professionals. Our team worked on building conversational agentic RAG engines that powers the company's core products (clinic coordination, finance assistant).

LG AI @ Seoul, Korea Sep 2023 - Feb 2024
Research Intern
I worked on an evaluation benchmark, InstructIR, for measuring user-aligned instance-wise instruction following ability of information retrieval at Language lab [7].

Kakao Enterprise @ Seongnam, Korea Jul 2020 - Oct 2020
Machine Learning Engineer Intern
I worked on knowledge graph embedding (KGE) to better represent complex relationship of large-scale in-house graph data. I also contributed to analyze large scale graph data with distributed system with Hadoop and Spark.

NAVER @ Seongnam, Korea Mar 2020 - Jun 2020
Software Engineer Intern
I worked on tools for entity linking engine and neural entity linking model for the search system.

Undergraduate Research Program, KOFAC @ Seoul, Korea Jan 2019 - Dec 2019
Undergraduate Researcher
I worked on a project for building a fake news detection system for news. Our team is grouped in multidisciplinary with social science, media, and industrial engineering major.

Republic of Korea Air Force @ Jinju, Korea — Sep 2015 - Sep 2017
Sergeant
I served as a training assistant for the Republic of Korea Air Force Training Wing.

Honors and Awards

- Best paper, NAACL 2025 [8]
- 1st Place, VALUE Challenge Retrieval Track, ICCV 2021 [1]
- Academic Excellence Scholarship, Sungkyunkwan University, Korea (2019)

Selected Papers

[10] Myeong Seok Oh, Dong-Yun Kim, **Hanseok Oh**, Chaean Kang, Joeun Kang, Xiaonan Wang, Hyunjung Park, Young Cheol Jung, and Hansaem Kim. "Subject-level Inference for Realistic Text Anonymization Evaluation". In: *ACL*. 2026.

[9] Hyunji Lee, Seunghyun Yoon, Yunjae Won, **Hanseok Oh**, Geewook Kim, Trung Bui, Franck Dernoncourt, Elias Stengel-Eskin, Mohit Bansal, and Minjoon Seo. "Instruction Tuning with and without Context: Behavioral Shifts and Downstream Impact". In: *EACL*. 2026.

[8] Seungone Kim, Juyoung Suk, Ji Yong Cho, Shayne Longpre, Chaeeun Kim, Dongkeun Yoon, Guijin Son, Yejin Cho, Sheikh Shafayat, Jinheon Baek, Suehyun Park, Hyeonbin Hwang, Jinkyung Jo, Hyowon Cho, Haebin Shin, Seongyun Lee, **Hanseok Oh**, Noah Lee, Namgyu Ho, Sejune Joo, Miyoung Ko, Yoonjoo Lee, Hyungjoo Chae, Jamin Shin, Joel Jang, Seonghyeon Ye, Bill Yuchen Lin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. "The BiGGen Bench: A Principled Benchmark for Fine-grained Evaluation of Language Models with Language Models". In: *NAACL*. 2025. **Best Paper Award**.

[7] **Hanseok Oh**, Hyunji Lee, Seonghyeon Ye, Haebin Shin, Hansol Jang, Changwook Jun, and Minjoon Seo. "INSTRUCTIR: A Benchmark for Instruction Following of Information Retrieval Models". In: *ACL workshop on Knowledge-Augmented NLP*. 2024. **Oral**.

[6] **Hanseok Oh**, Haebin Shin, Miyoung Ko, Hyunji Lee, and Minjoon Seo. "KTRL+ F: Knowledge-Augmented In-Document Search". In: *NAACL*. 2024.

[5] Yongrae Jo, Seongyun Lee, Aiden SJ Lee, Hyunji Lee, **Hanseok Oh**, and Minjoon Seo. "Zero-shot dense video captioning by jointly optimizing text and moment". In: *arXiv preprint*. 2023.

[4] Hyunji Lee, Jaeyoung Kim, Hoyeon Chang, **Hanseok Oh**, Sohee Yang, Vlad Karpukhin, Yi Lu, and Minjoon Seo. "Nonparametric Decoding for Generative Retrieval". In: *ACL Findings*. 2023.

[3] **Hanseok Oh**, Seojin Nam, and Yongjun Zhu. "Structured abstract summarization of scientific articles: Summarization using full-text section information". In: *JASIST* (2023).

[2] Hyunji Lee, Sohee Yang, **Hanseok Oh**, and Minjoon Seo. "Generative multi-hop retrieval". In: *EMNLP*. 2022. **Oral**.

[1] Aiden Seungjoon Lee, **Hanseok Oh**, and Minjoon Seo. "ViSeRet: A simple yet effective approach to moment retrieval via fine-grained video segmentation". In: *ICCV Workshop on Closing the Loop between Vision and Language (CLVL)*. 2021.

Services

- Reviewer: ARR, ACL, NAACL, EMNLP
- Sub-reviewer: NeurIPS, ECCV